

# MEP Drawing Interpretation

AI Engineering Intern · Suggested duration: 2 days

This exercise reflects a problem we work on directly: the automated interpretation of MEP (mechanical, electrical, and plumbing) construction drawings for enterprise clients. The task below is a deliberately narrow version of it. We are evaluating how you scope, reason, and build, rather than the finish of the final product.

## The task

Build a web application, runnable locally, in which a user uploads an MEP drawing and asks a question about it. The application answers using a vision-language model (VLM).

Two requirements apply to every answer:

- **Grounding.** The answer must show where on the drawing the information was found, with an indication of confidence.
- **Uncertainty.** Where the system is not confident, it should flag the item for human review rather than guess.

## Your drawings

| Drawing                    | Source & what it is                                                                                                                                                                                                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Drawing 1 (primary)</b> | <a href="#">RXR Welcome Center — HVAC Floor Plan (M-101)</a><br>A single HVAC floor plan with numerous ceiling diffusers and a symbol legend. The Drawing 1 questions below apply to this sheet.                                                             |
| <b>Drawing 2</b>           | <a href="#">Muswellbrook Depot — Mechanical Services</a><br>A separate mechanical set from a different source — a cover sheet with legends and floor-plan layouts. The Drawing 2 questions test whether your application generalises beyond the first sheet. |

## The questions your application must answer

Your application should answer all ten questions below, and the same application must work on both drawings without logic hard-coded to a single file. Every answer should indicate where on the drawing the information was found and flag anything uncertain. Where the correct answer is that an item is not present, the system should report that rather than return something that is not there.

### Drawing 1 — RXR Welcome Center HVAC plan

1. How many ceiling supply diffusers are shown on this floor, and where is each one located?
2. What return-air and transfer grilles are shown, and how are they distinguished from the supply diffusers?
3. List the exhaust fans on the sheet and their approximate locations.
4. What general notes apply to the diffusers and ductwork on this sheet — for example, anything specified for all diffusers or all duct runs?
5. Does the sheet contain any flag notes or keynotes indicating an unresolved field condition or coordination conflict to verify? If so, summarise each.

### Drawing 2 — Muswellbrook Mechanical Services

1. Using this drawing's legend, what is the difference between a supply air diffuser and a supply air swirl diffuser, and how many of each appear on the Upper Ground layout?
2. What mechanical equipment is shown across the set (e.g. fan coil units, VRF condenser, mechanical switchboard, MVHR unit), and on which level does each appear?
3. How many fire dampers are shown on the layouts, and where?
4. Decode the abbreviations defined on the cover sheet (e.g. FW, TD, CP, DG, UC) and locate one example of any one of them on a layout.
5. Distinguish refrigerant pipework from condensate pipework using the pipework legend, and describe where each runs.

## What to deliver

- **A runnable application** — we should be able to start it locally from your README with a single documented command.
- **The source repository** — your code, with the README described above.
- **A decision log (approximately two pages)**. Covering what you built and what you deliberately left out; the main technical choices and your reasons for them; where the system is unreliable and how you established that; and how you would extend the approach to large, multi-sheet drawing sets. Prose is sufficient; there is no need to build this part.
- **A 3–5 minute screen recording** demonstrating two or three of these questions end to end.

## Guidance

- We would rather see a smaller number of questions answered reliably than all ten attempted superficially. Prioritise accordingly, and record in your decision log what you chose not to do and why.
- You may use AI coding tools such as Claude Code, Cursor, or Copilot, and we assume you will. We assess how you direct these tools and the reasoning behind your decisions, not whether you wrote every line by hand.
- There is no single correct model or architecture. We are interested in your reasoning, including the alternatives you considered and set aside.

## What we assess

We are assessing four things: your ability to work independently, to scope the problem sensibly, to deliver an application that runs, and to reason clearly about where a VLM-based system performs well and where it fails. A limited solution accompanied by an accurate account of its failure modes is stronger than an ambitious one presented as complete.

## The debrief

After submission, you will have a short debrief with Vivek, our founder, to review your solution: the decisions you made, what you would change given more time, and a small live modification to the running application. Please be prepared to navigate and explain your own code during this session, as this matters more to us than the finish of the submission.

## Submission

Please send the repository link, decision log, and recording to [vivek@encraft.ai](mailto:vivek@encraft.ai) by 5<sup>th</sup> June 2026 midnight (end of day). If anything in this brief is unclear, you are welcome to ask.